Nuclear Inst. and Methods in Physics Research, A 988 (2021) 164830

Contents lists available at ScienceDirect

Nuclear Inst. and Methods in Physics Research, A

journal homepage: www.elsevier.com/locate/nima

Plasma focus neutron energy and anisotropy measurements using zirconium–beryllium pair activation detectors

S.V. Springham *, R. Verma, M.S.N. Zaw, R.S. Rawat, P. Lee, A. Talebitaher, J.H. Ang

Natural Sciences and Science Education, National Institute of Education, 637616, Singapore

ARTICLE INFO

Keywords:
Plasma focus
DD fusion
Neutron activation detector
Neutron energy
Anisotropy

ABSTRACT

Zirconium and beryllium fast-neutron activation detectors are used to investigate the fusion neutron emission from the NX3 Plasma Focus (PF) device operated in deuterium gas, with 7.2 kJ bank energy. Both Zr and Be activation cross-sections increase with energy, but with different trends, enabling an effective neutron energy $E_{\mathrm{n}}^{\mathrm{eff}}$ to be inferred from the Zr/Be count ratio. The relationship between $E_{\mathrm{n}}^{\mathrm{eff}}$ and Zr/Be count ratio is established by MCNP5 simulation. Compact Zr/Be detector pairs were positioned at 0° and 90° to the PF axis, permitting measurements of neutron yield, energy- and fluence-anisotropy to be made for each shot. Series of NX3 shots were performed for D_2 gas pressures ranging from 1.5 to 10 mbar. Typical effective neutron energies $E_{\mathrm{n}}^{\mathrm{eff}}$ for the 0° and 90° directions are found to be ~2.8 MeV and ~2.5 MeV, respectively. The highest neutron yields of ~10^9 neutron/shot were observed for 5 mbar D_2 gas pressure. Neutron fluence-anisotropy A_{nBe} for individual shots ranged from ~2.5 to ~4.5. The mean value $\langle A_{\mathrm{nBe}}\rangle$ exhibits a steady decline with increasing D_2 gas pressure. By contrast, neutron energy-anisotropy ΔE_{n} remains almost constant as the D_2 gas pressure is varied. The effect of blocking the forward D^+ ion beam with an obstacle plate positioned 6 cm in front of the anode tip is also studied. Marked reductions in both neutron yield and fluence-anisotropy are observed, whilst the effective neutron energy $E_{\mathrm{n}}^{\mathrm{eff}}$ increases slightly at both 0° and 90° directions. Fusion contributions from thermonuclear or gyrating-particle processes are found to be negligible. All results are completely consistent with a straightforward beam–target model of PF fusion.

1. Introduction

When operated with deuterium gas, Plasma Focus (PF) devices produce (~100 ns duration) bursts of neutrons with energies in the approximate range 2.1–3.1 MeV [1,2]. These neutrons originate from the $^2\mathrm{H}(d,n)^3\mathrm{He}$ fusion reaction: $^2_1\mathrm{H} + ^2_1\mathrm{H} \rightarrow ^3_2\mathrm{He} + ^1_0\mathrm{n}$ with $Q = +3.27$ MeV. Thermonuclear fusion produces neutrons with energies close to 2.45 MeV. In the PF, fusion occurs in the dense plasma pinch column (formed at the end of the plasma compression phase), and also in a conical region downstream of the pinch column due to an intense forwardly-directed beam of energetic D^+ ions (accelerated within the pinch) bombarding deuterons in the cold filling gas [1–3], and it is known that neutron energy varies from shot to shot and with the angle θ (wrt the forward PF axis) [4]. For a beam–target DD fusion reaction, the solution of the kinematic Q-equation [5] gives the energy E_n of neutrons emerging at angle θ_n as

$$E_n = A + E_d\left(\cos^2\theta_n + 1\right)/4 + \sqrt{E_d\left(\cos^2\theta_n + 2\right)/16 + B}\sqrt{E_d}\cos\theta_n \quad (1)$$

where E_d is the energy of the incident (beam) deuteron, and the constants are: $A = 2.451$ MeV and $B = 1.226$ MeV. The relationship between θ_n and E_n is plotted in Fig. 1(b) for several values of E_d. Shown in Fig. 1(a) are the fast neutron activation cross-sections for ^{90}Zr and Be: $^{90}\mathrm{Zr}(n, n'_3 + n'_4)^{90\mathrm{m}}\mathrm{Zr} \rightarrow \sigma^{\mathrm{a}}_{\mathrm{Zr}}(E_{\mathrm{n}})$ and $^9\mathrm{Be}(n,\alpha)^6\mathrm{He} \rightarrow \sigma^{\mathrm{a}}_{\mathrm{Be}}(E_{\mathrm{n}})$, respectively. Experimental studies have reached differing conclusions on the deuteron energy range responsible for the bulk of the PF neutron yield: e.g. E_d = 30–350 keV in [2]; E_d = 30–60 keV in [3]; and E_d = 50–100 keV [6].

Neutron fluence and energy measurements can help to elucidate the mechanism and spatial extent of fusion in the PF. However, it is especially challenging to measure the energy spectrum (or even the mean energy) of the emitted neutrons. The two principal methods previously employed are time-of-flight (TOF) [1,2,4,6,8–10] and proton-recoil in nuclear emulsion [11–14]. The nuclear emulsion technique is exceptionally time consuming, and best suited to measuring a cumulative many-shot energy spectrum. It would be an onerous task to use nuclear emulsion to measure many individual PF shots. The emulsion technique has other drawbacks: low neutron detection efficiency (due to the small detector mass, and limited H atom content), and it requires that the direction of the incoming neutron be known, which necessitates a point-like neutron source. For example, in Ref. [12] emulsion was exposed for 169 shots (0.5–2 × 10^{10} n/shot) at 28 cm distance from the

* Corresponding author.
E-mail address: stuart.springham@nie.edu.sg (S.V. Springham).

https://doi.org/10.1016/j.nima.2020.164830
Received 31 January 2020; Received in revised form 5 October 2020; Accepted 28 October 2020
Available online 4 November 2020
0168-9002/

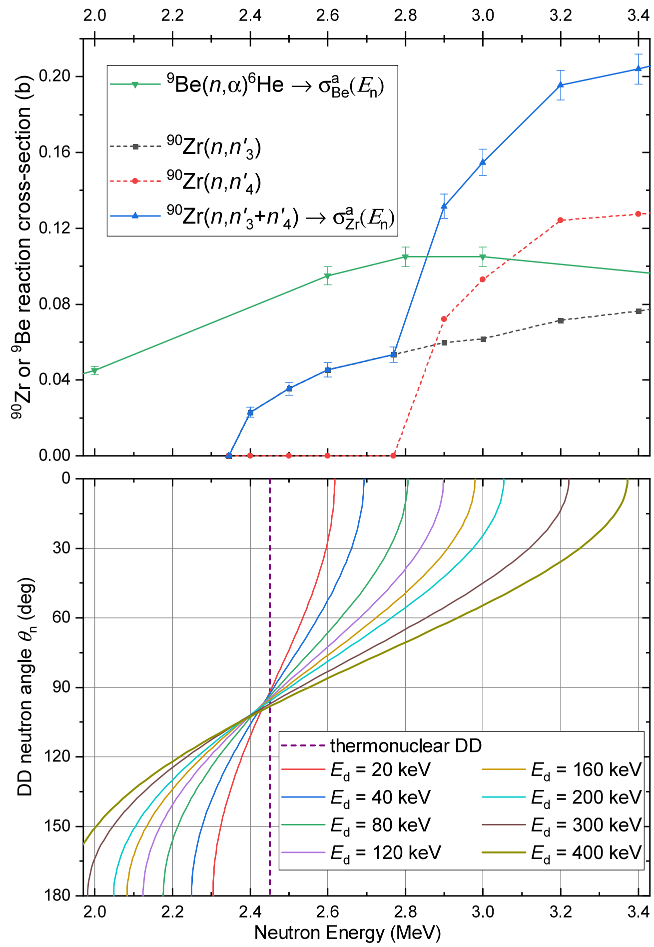

Fig. 1. (a): plots of the reaction cross-sections (data from [7]) for ^{9}Be$(n,\alpha)^6$He and ^{90}Zr $(n,n')^{90m}$Zr leading to activated product nuclei. n'_3 and n'_4 correspond to excitation of the 2.319 and 2.739 MeV levels of ^{90}Zr, respectively, which both result in activation. (b): Plots of DD fusion neutron energy as a function of neutron emission angle θ_n, for various deuteron bombarding energies E_d. Thermonuclear fusion neutron energy (2.45 MeV) is also indicated.

anode tip. While in Ref. [11] the nuclear emulsion is placed rather close (∼5 cm) to the pinch, making the neutron point-source assumption quite dubious. For TOF, the ∼100 ns burst duration and relatively narrow neutron energy range necessitates the use of long flight-paths, large PF devices with high neutron yields, and large neutron detectors. For example, for the 100 kJ PF with $\sim 10^9$ neutron yield in Ref. [1], flight paths of 120 m and plastic scintillator detectors (vol. ∼21 litre) were used at 0° and 90°, simultaneously, to attain $\Delta E_n < 0.1$ MeV energy resolution. A long flight path at 0° usually requires that the PF axis be horizontal. (Whereas both the NX2 and NX3 axes are vertical). Such arrangements are only feasible for large experimental facilities. Even when such spacious TOF facilities are available, neutron scattering from the vacuum chamber, concrete floor and walls, etc., complicates the extraction of accurate energy spectra.

A completely different approach to measuring PF neutron energy is developed in this work. Utilizing pairs of zirconium and beryllium Fast Neutron Activation (FNA) detectors enables a (quasi-average) ‘effective neutron energy’ to be determined for each shot of the NX3 PF device, with a typical neutron yield of $\sim 5 \times 10^8$/shot. The Zr/Be detector Pairs (ZBP) are sufficiently compact that two of them can be used simultaneously at $\theta = 0°\&90°$. For the sake of brevity, the detector pairs at these angles will be referred to as ZBP-0° and ZBP-90°. Conceptually, the present technique is related to the spectrum unfolding technique (SUT) which uses n single-element activation foils ($j = 1, \dots, n$) for which neutron-induced activation reactions occur with different threshold energies, E_j^{th}. This standard technique is described in Ref. [15]. SUT is typically used to unfold wide energy spectra (from thermal to ∼15 MeV) in very high flux environments, such as neutron spallation sources. The activation half-lives (τ_j) used may range from minutes to months. Unlike SUT, the present technique applies to a narrow energy spectrum, specifically non-thermonuclear DD fusion; the neutron flux is low and the half-lives are short ($\tau_j < 1$ s). So, despite conceptual similarities, there are large practical differences between the present technique and SUT. For situations where the neutron burst duration Δt_b is much shorter than the activation half-lives ($\Delta t_b \ll \tau_j$), the expected sum of secondary radiation counts S_j in the jth detector is

$$S_j = \left(e^{-\lambda_j t_s} - e^{-\lambda_j t_e}\right) \int_{E_j^{\text{th}}}^{E_{\max}} N_j \sigma_j^{\text{a}}(E)\, \Phi_j(E)\, b_j \eta_j \, dE \tag{2}$$

where t_s and t_e are count start and end times, respectively, for the burst occurring at $t = 0$. Naturally $0 < t_s < t_e$. For the jth activation: $\lambda_j = \ln 2/\tau_j$ is the decay constant for the activated nuclide; N_j is the number of parent nuclei present, $\sigma_j^{\text{a}}(E)$ is the activation cross-section; $\Phi_j(E)$ is the energy-dependent neutron flux at the jth target position; b_j is the branching-ratio for the emitted secondary radiation; and η_j is the overall secondary radiation detection efficiency (i.e. integrated over the energy range of the secondary radiation, and subsuming the geometrical efficiency). In this work only two activation targets are used, and both branching ratios are $b_j = 1.0$. Then the expected count ratio $\mathfrak{R}_{\text{ZrBe}}$ between Zr and Be activation detectors is

$$\begin{aligned}\mathfrak{R}_{\text{ZrBe}} &= \frac{S_{\text{Zr}}}{S_{\text{Be}}} \\ &= \frac{0.5145 N_{\text{Zr}} \eta_\gamma \left(e^{-\lambda_{\text{Zr}} t_s} - e^{-\lambda_{\text{Zr}} t_e}\right)}{N_{\text{Be}} \eta_\beta \left(e^{-\lambda_{\text{Be}} t_s} - e^{-\lambda_{\text{Be}} t_e}\right)} \frac{\int_{2.35\,\text{MeV}}^{E_{\max}} \sigma_{\text{Zr}}^{\text{a}}(E)\, \Phi_{\text{Zr}}(E)\, dE}{\int_{0.67\,\text{MeV}}^{E_{\max}} \sigma_{\text{Be}}^{\text{a}}(E)\, \Phi_{\text{Be}}(E)\, dE}\end{aligned} \tag{3}$$

Eq. (3) applies to the thin targets case, where neutron attenuation and multiple-scattering are negligible. ^{90}Zr (51.45%) and ^{9}Be (100%) are the natural isotopic abundances of the target nuclei. N_{Zr} and N_{Be} are the numbers of Zr and Be atoms present in the respective targets. Strictly, the ^{90}Zr activation cross-section ought to be denoted $\sigma_{\text{Zr90}}^{\text{a}}(E_n)$, but here (and in Fig. 1) it is shortened to $\sigma_{\text{Zr}}^{\text{a}}(E_n)$. η_γ and η_β are the overall detection efficiencies for ^{90}Zr γ-rays and ^{6}He β-particles for the secondary radiation detectors used in this work. The lower integration limit is the threshold energy for the associated neutron-induced reaction. For the simplifying case of an isotropic point source of mono-energetic neutrons, energy E_n, the $\mathfrak{R}_{\text{ZrBe}}$ ratio becomes

$$\begin{aligned}\mathfrak{R}_{\text{ZrBe}} &= \frac{S_{\text{Zr}}}{S_{\text{Be}}} = \frac{0.5145 N_{\text{Zr}} \eta_\gamma \left(e^{-\lambda_{\text{Zr}} t_s} - e^{-\lambda_{\text{Zr}} t_e}\right)}{N_{\text{Be}} \eta_\beta \left(e^{-\lambda_{\text{Be}} t_s} - e^{-\lambda_{\text{Be}} t_e}\right)} \frac{\sigma_{\text{Zr}}^{\text{a}}(E_n)\, R_{\text{Be}}^2}{\sigma_{\text{Be}}^{\text{a}}(E_n)\, R_{\text{Zr}}^2} \\ &= \frac{\left(e^{-\lambda_{\text{Zr}} t_s} - e^{-\lambda_{\text{Zr}} t_e}\right)}{\left(e^{-\lambda_{\text{Be}} t_s} - e^{-\lambda_{\text{Be}} t_e}\right)} \frac{\text{P}_{\text{Zr}}(E_n)}{\text{P}_{\text{Be}}(E_n)} = \frac{\left(e^{-\lambda_{\text{Zr}} t_s} - e^{-\lambda_{\text{Zr}} t_e}\right)}{\left(e^{-\lambda_{\text{Be}} t_s} - e^{-\lambda_{\text{Be}} t_e}\right)} G(E_n)\end{aligned} \tag{4}$$

where R_{Zr} and R_{Be} are the radii of the target mid-points from the neutron source. For a fixed arrangement of activation detectors, many quantities can be gathered into a response function [16]: $\text{P}_{\text{Zr}}(E_n) = 0.5145 N_{\text{Zr}} \eta_\gamma \sigma_{\text{Zr}}^{\text{a}}(E_n)/4\pi R_{\text{Zr}}^2$ and $\text{P}_{\text{Be}}(E_n) = N_{\text{Be}} \eta_\beta \sigma_{\text{Be}}^{\text{a}}(E_n)/4\pi R_{\text{Be}}^2$. Let $G(E_n) = \text{P}_{\text{Zr}}(E_n)/\text{P}_{\text{Be}}(E_n)$. However, because fast-neutron activation cross-sections are low, thick targets must be used and both multiple-scattering of neutrons and gamma-ray self-shielding become significant effects. In which case $\text{P}_{\text{Zr}}(E_n)$ and $\text{P}_{\text{Be}}(E_n)$ are best obtained by Monte Carlo simulation. Replacing the expectation count ratio $S_{\text{Zr}}/S_{\text{Be}}$ in Eq. (4) with the experimentally measured ‘uncorrected’ count ratio $C_{\text{Zr}}^{\text{u}}/C_{\text{Be}}^{\text{u}}$, the effective neutron energy $E_{\text{n}}^{\text{eff}}$ is defined by

$$E_{\text{n}}^{\text{eff}} = G'\left(\frac{e^{-\lambda_{\text{Be}} t_s} - e^{-\lambda_{\text{Be}} t_e}}{e^{-\lambda_{\text{Zr}} t_s} - e^{-\lambda_{\text{Zr}} t_e}} \times \frac{C_{\text{Zr}}^{\text{u}}}{C_{\text{Be}}^{\text{u}}}\right) = G'\left(\frac{C_{\text{Zr}}^{\text{c}}}{C_{\text{Be}}^{\text{c}}}\right) \tag{5}$$

where $G'(x)$ is the inverse function of $G(E)$. Also, C_{Zr}^{c} and C_{Be}^{c} are the corrected net counts discussed in Section 3.

In our previous work [17], with the NX2 PF device operated at 1.6 kJ stored energy, two beryllium FNA detectors were positioned at 0° and 90° with respect to the PF axis to investigate neutron yield and fluence-anisotropy. For some shots, a Pyrex plate was used to obstruct the D^+ ion-beam and suppress its fusion contribution: two pinch-to-obstacle distances were investigated (3.0 and 6.0 cm). It was found that the average neutron fluence-anisotropy $\langle A_{nBe} \rangle$ decreased as the Pyrex obstacle was moved closer to the PF anode. Our interpretation was that this decreasing $\langle A_{nBe} \rangle$ occurs for two reasons: (i) shifting of the neutron emission centre-of-gravity towards the anode tip (and therefore further from the 0° detector), and (ii) the fusion originating within the pinch has intrinsically lower anisotropy than the fusion occurring in the forward conical region. (See Fig. 2 of Ref. [2]). Reason (ii) was viewed in terms of the Gyrating-Particle-Model [18], where D^+ ions in the pinch column are trapped into gyrating-particle orbits by its azimuthal magnetic field B_ϕ. Two major differences from our work in Ref. [17] are: (I) The present study employs the higher capacitor-bank energy (20 kJ) NX3 PF device. The neutron yield of the NX3 is about an order of magnitude greater than that of the NX2, ($\sim 10^9$ vs. $\sim 10^8$ neutron/shot). (II) Newly developed zirconium FNA detectors are paired with the beryllium FNA detectors (as used in [17]). The FNA pairing permits effective neutron energies to be inferred at $\theta = 0°\&90°$. A minor difference is that only one anode-to-obstacle distance is used here, while two distances were studied in [17].

The operation and calibration of the beryllium FNA detector is explained in Ref. [16]. Fast neutrons induce $^9Be(n,\alpha)^6He$ reactions in a 5 mm thick Be metal plate producing a population of ^6He nuclei which undergo β-decay with 807 ms half-life. Xenon filled proportional counters, placed either side of the Be plate, detect approximately 40% of all the ^6He decay β-particles. Beryllium FNA detectors of a different design are being developed for other pulsed fusion devices [19]. On the other hand, FNA of zirconium is due to inelastic scattering $^{90}Zr\,(n,n')^{90m}Zr$ producing a population of metastable ^{90m}Zr nuclei. The de-excitation γ-rays are detected by a BGO scintillator placed in close proximity to the activated Zr target. As shown in Fig. 1(a), the $^9Be(n,\alpha)^6He$ reaction cross-section increases slowly with neutron energy from 2.0 to 3.0 MeV [20,21], while the $^{90}Zr\,(n,n')^{90m}Zr$ cross-section has a threshold energy of 2.345 MeV and increases rapidly between 2.5 and 3.2 MeV [21,22]. It is these differing cross-section trends which permit an effective neutron energy E_n^{eff} to be inferred from the Zr/Be count ratio for each PF shot. The activation reactions and secondary radiations are illustrated below.

$$n + {}^{90}Zr \rightarrow {}^{90m}Zr\,(t_{1/2} = 809\text{ ms}) \rightarrow + n' \rightarrow {}^{90}Zr + \gamma\,(2.319 \text{ or } 2.189\text{ MeV})$$

$$n + {}^{9}Be \rightarrow {}^{6}He\,(t_{1/2} = 807\text{ ms}) \rightarrow + {}^4He \rightarrow {}^6Li + \beta \quad Q_{\beta\text{-}} = 3.51\text{ MeV}$$

A coincidental, but useful, aspect of these activations is the very similar (0.3% different) half-lives of ^{90m}Zr and ^6He, which are 809.2 ms and 806.7 ms respectively. This makes it convenient to use the same 3.0 s interval in the pulse processing electronics to acquire the counts for all four FNA detectors: two zirconium + two beryllium. It is worth mentioning that both Be and Zr have exceptionally low thermal neutron capture (n,γ) cross-sections, and the thermal capture products are stable or very long-lived. Hence, the response of Be and Zr activation detectors is entirely negligible for thermal neutrons. In fact, it is negligible for $E_n < 1$ MeV.

2. Experimental setup

2.1. NX3 plasma focus device

The design and optimization of the NX3 plasma focus are fully described in Ref. [23]. The NX3 electrode configuration comprises a cylindrical stainless steel anode (26 mm radius, 126 mm length), and a squirrel cage cathode of six copper rods (12 mm diameter), uniformly spaced on a coaxial circle of 56 mm radius. These dimensions make it a Mather-type PF. The device is energized by a 100 μF, 20 kJ, modular capacitor bank comprising 8 parallel capacitors: 12.5 μF, 20 kV, 80% reversal, 150 kA, with < 40 nH inductance, from Aerovox Corporation. Triggering electronics and fast pseudo-spark switches (PSS) (150 kA/20 kV hold-off) from M/s Pulsed Technologies Ltd., are installed on each capacitor for rapid energy transfer to the NX3 PF load. The specified inductance and jitter of the PSS are 20 nH and 3 ns, respectively. The design discharge quarter-time period for 5 mbar D_2 gas pressure is 3 μs. The stainless steel NX3 vacuum chamber has outer diameter 25.6 cm, height 40.0 cm, and the distance from the PF axis to the outside surface of the glass window (where ZBP-90° resides) is 28.0 cm.

A charging voltage of 12 kV (stored energy = 7.2 kJ, and peak discharge current ∼ 600 kA) was used throughout these experiments. The D_2 gas pressures used were: 1.5 mbar, and 2 to 10 mbar in steps of 1 mbar (giving 10 different pressures). Fig. 2 shows a scaled schematic diagram of the NX3 PF vacuum chamber, together with the compact arrangement of Zr/Be FNA detector-pairs. The radial distances from the centre of the anode tip (and centre of plasma pinch) to the mid-points of the fast neutron activated materials are as follows: $R_{Be0} = 296$ mm, $R_{Be90} = 318$ mm, $R_{Zr0} = 320$ mm, and $R_{Zr90} = 342$ mm. At the plasma pinch, the solid angles subtended by the Be targets are 0.237 and 0.207 sr for 0° and 90° detectors, respectively; less than 2% of the whole sphere. The Zr targets subtended smaller solid angles.

2.2. Data acquisition system

A block diagram of the data acquisition system is shown in Fig. 3. The pulse processing electronics and computer were located inside a large screen-cage to minimize interference from the discharge-produced transient RF noise. The counting interval $\Delta\tau = 3.0$ s $\cong$ 3.7 half-lives is implemented in the pulse processing electronics to count both ^{90m}Zr emitted γ-rays (BGO) and ^6He emitted β-particles (proportional counters). Four Ortec 773 NIM scalers, enabled by the 3.0 s gate-pulse, accumulate the gross count for each of the four activation detectors. Additionally, one Multi-Channel Analyser (Ortec pci-MCA card and MAESTRO software) was operated in Pulse-Height-Analysis (PHA) mode. The PHA is also gated by the 3.0 s pulse; it accumulates the γ-ray energy spectrum from the BGO scintillation detector of either (0° or 90°) of the Zr activation detectors.

A Rogowski coil encircled one capacitor cable near its connection to the anode collector plate, providing a signal proportional to the discharge current derivative (dI/dt). This signal is recorded by a Yokogawa DL9140 oscilloscope (1 GHz bandwidth, 5 G sample/s). The oscilloscope is set to trigger on the rising edge of the (dI/dt) signal, and the 3.0 s gate-pulse is generated from the oscilloscope's Trigger-Out pulse delayed by 10 ms. Hence, this gate-pulse enables the 4 scalers, and one PHA over the time period $t = 0.01$ to 3.01 s relative to the start of the PF discharge. The 10 ms delay allows various forms of interference (hard x-rays, discharge EM noise, and prompt γ-rays from neutron captures) to die away before the $\Delta\tau = 3.0$ s counting interval begins.

2.3. Calibration procedures and MCNP5 simulations

Operation of the beryllium FNA detectors, and details of MCNP5 Monte Carlo simulations required for their calibration are fully described in Ref. [16]. Turning to the zirconium activation detectors, the de-excitation of metastable ^{90m}Zr nuclei to the ground state occurs via the emission of either a 2.319 MeV (82%) or 2.189 MeV (18%) energy γ-ray [21]. The signal train for the γ-ray spectroscopy system comprised: scintillator, photomultiplier, Ortec 113 pre-amp, Ortec 590A

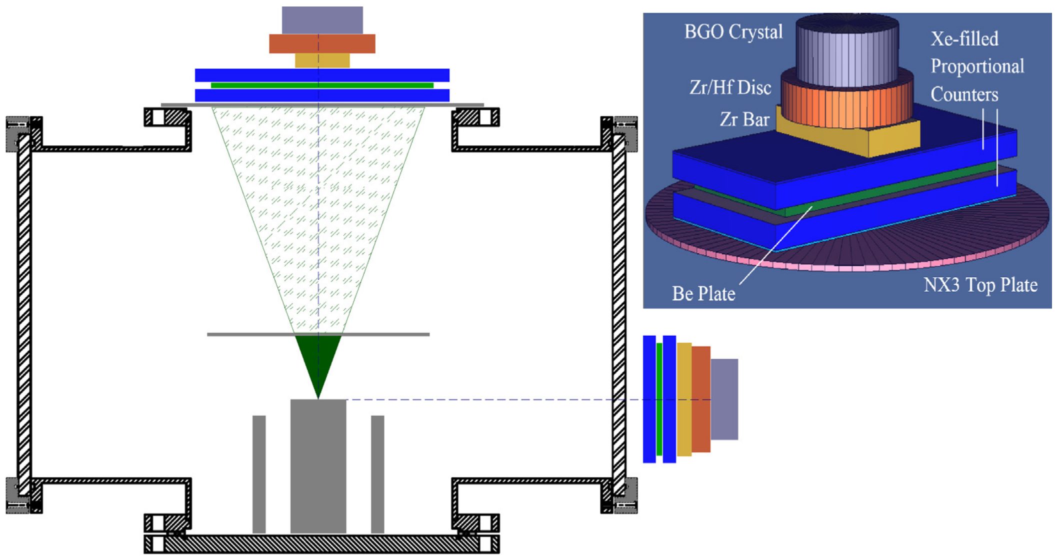

Fig. 2. Drawn to scale schematic of the NX3 PF vacuum chamber, electrodes, and Zr/Be fast-neutron activation detector pairs positioned at 0° and 90° with respect to the PF axis. An example, 20° half-angle deuteron beam emitted from the pinch is shown with the obstacle plate in position (OP case). For the w/o-OP case the conical deuteron beam extends to the top plate (i.e. shaded frustum contributes to fusion). Inset shows the 3D rendered geometry for MCNP5 simulation of the Zr/Be detector-pair (ZBP-0°) on the NX3 PF axis.

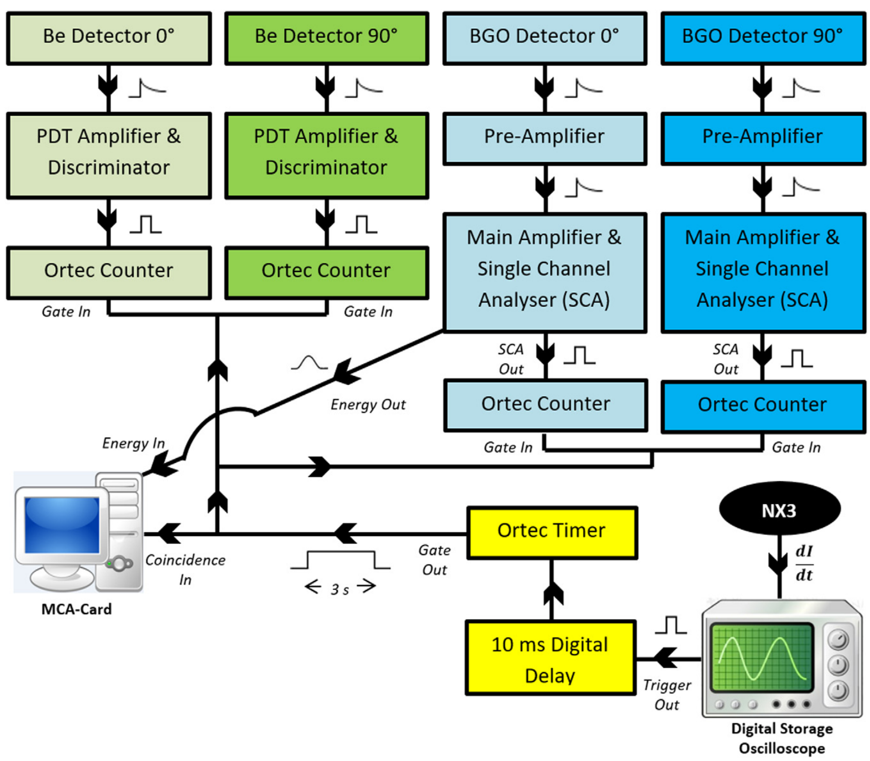

Fig. 3. Block diagram of the pulse processing electronics and data acquisition system.

amp & Single-Channel-Analyser (SCA). Energy calibration was performed using standard γ-ray lines from radioactive sources: ^{22}Na (E_γ = 0.511 and 1.275 MeV), and thorium-hydroxide (E_γ = $\cdots$, 2.61 MeV). The 2.61 MeV γ-rays from ^{208}Tl is especially useful for obtaining the Gaussian Energy Broadening (GEB) parameters (FT8 card) used within MCNP5 simulations of the expected BGO energy spectrum (F8 tally) obtained for ^{90m}Zr γ-rays emitted throughout the relatively thick Zr target. Suitable GEB parameters for our BGO detectors were found to be $b = 0.09368$ MeV$^{1/2}$ with $a = c = 0$, and hence simply FWHM $= b\sqrt{E}$. The SCA energy-windows were adjusted so that output pulses were generated only for BGO detector charge pulses corresponding to the deposited energy range 0.40 – 2.60 MeV. This SCA energy window accepts the full width of the 2.189 and 2.319 MeV photopeaks, along with their escape peaks, and a large part of their Compton continuum. It eliminates counting of lower energy γ-rays (E_γ < 400 keV) which constitute the majority of the background radiation.

The scalers are zeroed after each PF shot, but the PHA is not. Instead the PHA accumulates its spectrum over many 3 s gate intervals (i.e. for a series of shots). Fig. 4 shows BGO γ-ray foreground and background PHA energy spectra (foreground accumulated over 112 consecutive PF shots). The largest peak comprises the two unresolved γ-ray lines at 2.319 and 2.189 MeV; the intensity of the first line is greater by

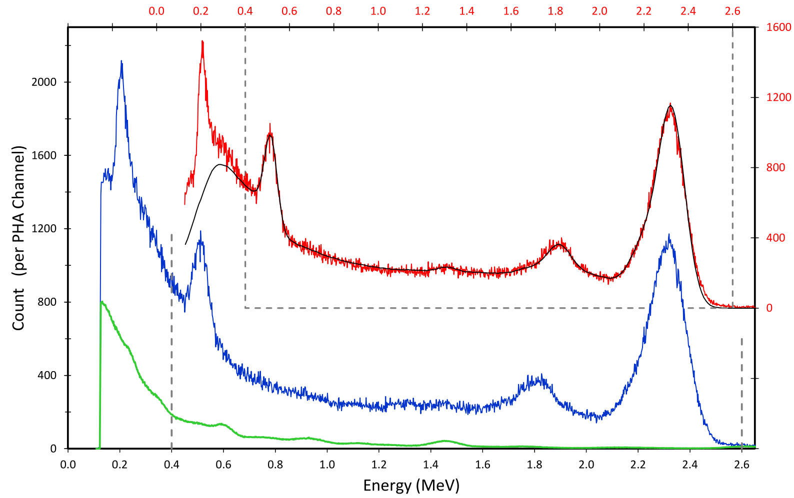

Fig. 4. (Blue) Pulse Height Analyser (PHA) spectrum from Zirconium-activation BGO detector at 0° to PF axis, accumulated over 7 series of shots, 16 shots/series, 3.0 s gate-interval/shot, giving a cumulative live-time = 336 s for 112 consecutive PF shots. (Green) PHA spectrum of Lab background radiation for live-time = 160 min = 9600 s, but scaled down to 336 s live-time. On secondary axes: (Red) Net (background subtracted) PHA spectrum, and (Black) MCNP5 simulated BGO energy spectrum for ^{90m}Zr emitted γ-rays. Grey dashed box represents SCA energy window. Numbers of counts per shot lying within the SCA energy window are: Gross = 3759/shot, Background = 353/shot, Net = 3406/shot. The PHA bin width is 1.93 keV.

a factor 4.6, so the combined peak is near 2.319 MeV. The single escape peak, due to pair-production (PP) within the BGO crystal, is evident at 1.81 MeV. The positron-annihilation peak at 0.511 MeV results mainly from PP in the zirconium target. Less expected is the peak just above 200 keV which is due to activation of the Hf content (2.3% by weight) of the Zr/Hf disc, via inelastic neutron scattering and thermal neutron capture: ^{179}Hf $(n,n')^{179m}$Hf and ^{178}Hf $(n,\gamma)^{179m}$Hf, respectively. Metastable ^{179m}Hf decays with an 18.7 s half-life, emitting a 0.214 MeV γ-ray [24]. This ^{179m}Hf peaks lies outside the SCA energy window, and so is not counted. PHA acquisition of BGO energy spectra provided confirmation that the zirconium FNA detectors were operating properly. This check was performed periodically, in turn, for each of the two BGO detectors. The respective amplifier gain settings were adjusted so that the same energy calibration, at the PHA input, was applicable to both BGO detectors. The good agreement between the net BGO energy spectrum (within the gate interval) and the MCNP5 simulated spectrum indicates that ^{90m}Zr γ-ray pulses far outnumber those due to activation γ-rays from room materials, and therefore the associated scaler count should be reliable. In our previous work Ref. [16] performed in the same Lab, the activation of aluminium (^{28}Al, 135 s, E_γ = 1.78 MeV) was identified as a main interfering activation. The photopeak for ^{28}Al γ-rays would almost coincide with the ^{90m}Zr single escape peak in the PHA spectrum, in Fig. 4. But the closeness of the fit to the simulated spectrum suggests that the number of ^{28}Al γ-rays is not significant in the present case.

MCNP5 simulations were performed to determine what fraction of ^{90m}Zr γ-ray de-excitations in the Zr target produce a BGO pulse lying within the SCA energy-window (and hence counted by the scaler). An MCNP 3D rendered graphic of ZBP-0° sitting on the top-plate of the NX3 vacuum chamber is shown in Fig. 2. A thin (3.0 mm) stainless steel top-plate was used in order to minimize the number of neutrons it scattered. Our zirconium metal targets were, for reasons of economy, bought in two different forms: Zr/Hf (sputtering target) discs (97.7% Zr by weight) (diameter 100 mm, thickness 18.1 mm), and '1 lb zirconium bullion bar' (99.2% Zr) (approx. dimensions: 52.5×107×12.5 mm). The masses of pure zirconium and beryllium for each of the ZBP-0°/90° detectors is m_{Zr} = 1348 g and m_{Be} = 206 g. The order of the ZBP elements (moving radially away from the pinch) is: Xe-PC, Be plate, Xe-PC, Zr bar, Zr/Hf disc, BGO crystal. The arrangement is symmetrical around the axis of the BGO crystal. ZBP-0° is on the PF axis, and ZBP-90° is level with the position of the PF anode tip.

Attenuation of the arriving neutrons and emitted ^{90m}Zr γ-rays in the Zr target is an important consideration. It is calculated, from Ref. [25] data, that the Half-Value Layer (HVL) for 2.319 MeV γ-rays in Zr is 2.73 cm (with 91% of the initial interactions being Compton scattering). Also, from Ref. [21] data, the 2.5 MeV neutron-elastic-scattering HVL is 3.8 cm in Zr. Hence, despite the 3.06 cm thickness, of the Zr target plate and disc combined, neutron and γ-ray attenuation/self-shielding are not dominant effects.

MCNP5 simulations of neutrons and γ-ray interactions are used to find the Zr and Be detector response functions $\mathrm{P}_{\mathrm{Zr}}\left(E_n\right)$ and $\mathrm{P}_{\mathrm{Be}}\left(E_n\right)$, and hence also $G(E)$, for E = 2.3 to 3.6 MeV at 0.1 MeV intervals. In these simulations, an isotropic mono-energetic neutron source is positioned at the centre of the anode tip. Included in the simulation are: neutron energy, source/chamber/detector geometry, material compositions and masses, γ-ray energies and branching ratios, ^{6}He β-particle energy spectrum, and Gaussian energy broadening (simulating the BGO energy resolution). Simulation results for $G'\left(C^{\mathrm{c}}_{\mathrm{Zr}}/C^{\mathrm{c}}_{\mathrm{Be}}\right)$ are plotted in Fig. 5. The mono-energetic (blue) curve is the basis for inferring the effective neutron energy $E^{\mathrm{eff}}_{\mathrm{n}}$. The slope-change near 2.8 MeV is due to excitation of the ^{90}Zr level at 2.739 MeV, which rapidly (<1 ns) transitions to the 2.319 MeV level ^{90m}Zr [21]. Hence this route also contributes to the population of ^{90m}Zr nuclei (i.e. activation). The threshold energies for excitation of each level is simply $(91/90)E_x$, where E_x is the level energy. It is worth considering how this $G'\left(C^{\mathrm{c}}_{\mathrm{Zr}}/C^{\mathrm{c}}_{\mathrm{Be}}\right)$ curve will be affected for a single Gaussian peaked neutron energy distribution rather than a mono-energetic source. The MCNP5 simulations were repeated for Gaussian peak neutron energy distributions with FWHM = 200&400 keV (Std.Dev. 85 and 170 keV, respectively). These plots are also shown in Fig. 5, but only the mono-energetic curve was used to calculate $E^{\mathrm{eff}}_{\mathrm{n}}$ for subsequent plots and tables.

The experimental uncertainties in the activation cross-sections $\sigma^{\mathrm{a}}_{\mathrm{Zr}}(E_{\mathrm{n}})$ and $\sigma^{\mathrm{a}}_{\mathrm{Be}}(E_{\mathrm{n}})$, shown in Fig. 1(a), inherently apply to the cross-section data used by MCNP5. Consequently, systematic uncertainties apply to the shape of the mono-energetic curve in Fig. 5. Mainly, this is likely to shift both θ = 0°&90° $E^{\mathrm{eff}}_{\mathrm{n}}$ values higher or lower, but the trends observed in $E^{\mathrm{eff}}_{\mathrm{n}}$ – from shot-to-shot, and `w/o-OP` vs. `OP` case, etc. – should remain the same. Ideally, future cross-sections measurements, similar to [24], could be performed with Zr and Be targets together, so that $\sigma^{\mathrm{a}}_{\mathrm{Zr}}(E_{\mathrm{n}})$ and $\sigma^{\mathrm{a}}_{\mathrm{Be}}(E_{\mathrm{n}})$ are measured simultaneously. Then due to the near identical half-lives involved, most sources of systematic error should cancel, and the ratio-function $(\sigma^{\mathrm{a}}_{\mathrm{Zr}}/\sigma^{\mathrm{a}}_{\mathrm{Be}})(E_{\mathrm{n}})$ would be more precisely determined than is achievable by separate cross-section measurements. A more direct approach would be to energy calibrate a specific design of Zr/Be FNA detector-pair using an accelerator source, again similar to Ref. [22].

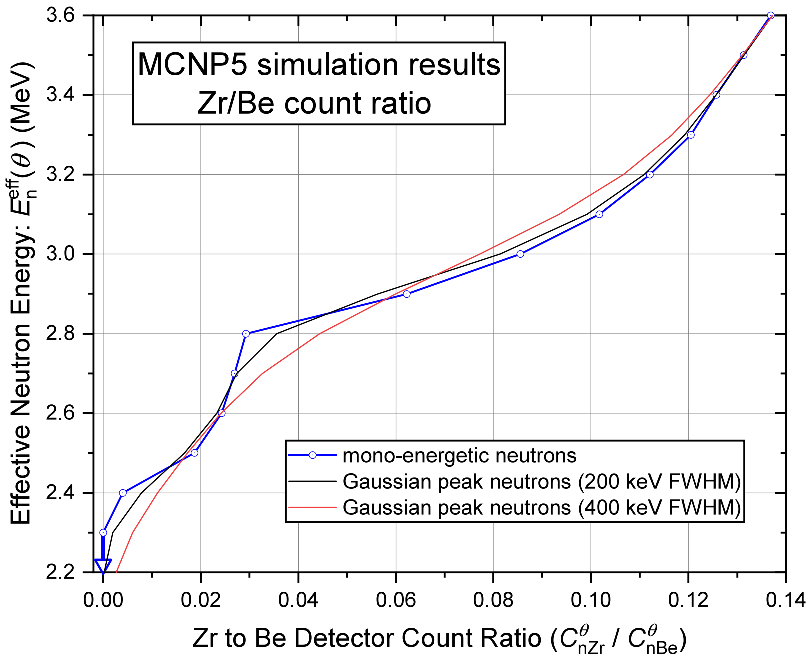

Fig. 5. Graph of MCNP5 simulation results for effective neutron energy E_n^{eff} versus Zr to Be detector count ratio $C^{\theta}_{nZr}/C^{\theta}_{nBe}$. For each shot: $E_n^{eff}(0°)$ and $E_n^{eff}(90°)$ are is obtained by interpolation from this plot. Simulated neutron source (at pinch): blue curve, mono-energetic; black curve, Gaussian peak (200 keV FWHM); red curve, Gaussian peak (400 keV FWHM).

2.4. Zirconium and beryllium fast-neutron activation detectors: operational details

Fig. 2 shows a schematic diagram of the NX3 vacuum chamber with Zr and Be FNA detector-pairs (drawn to scale) positioned immediately adjacent to it, at 0° and 90° orientations. ^{90m}Zr γ-rays were measured by Scionix (model 76B25/3M-E1) BGO scintillation detectors: 7.62 cm diameter, 2.54 cm thick, within a 0.5 mm aluminium housing. A mu-metal shield and built-in voltage divider complete the detector. Both BGO detectors were operated at 1200 V. BGO ($Bi_4Ge_3O_{12}$) has high values of effective atomic number (Z_{eff} = 74) and density (7.13g/cm^3) making it an efficient absorber of 2.319 and 2.189 MeV γ-rays from ^{90m}Zr de-excitations. The elemental constituents of BGO suffer negligible activation by fast or slow neutrons, and as a scintillator, BGO exhibits very low afterglow which permits the detector to recover from the (∼100 ns) burst of bremsstrahlung x-rays emitted during the shot. X-ray shielding of the BGO crystal by the zirconium target also helps in this respect. In this experiment, these characteristics of BGO far outweigh its modest energy resolution (due to its low light yield). Using data from [25], the HVL for the 2.319 MeV γ-ray in the (2.54 cm thick) BGO crystal is 2.28 cm. The overall detection efficiency of the ^{90m}Zr γ-ray events (within the SCA energy window) for the thick Zr target plus BGO detector is 9.3%.

For each beryllium FNA detector, the charge pulses from the Xe-PCs are processed by a PDT10A amplifier/discriminator units and for events above the 2.5 keV threshold a 200 ns TTL pulse is output. The PDT10A unit also extends the bias to the Xe-PCs, which were operated with 1500 V bias. The efficiency of ^{6}He β-particle detection is 42.6%.

It is apparent from Fig. 2 that the Zr detector responds to neutrons that are transmitted through the Be detector. A simple attenuation calculation shows that ∼85% of the DD neutrons are transmitted unscattered by the beryllium plate ($\sigma_n^{tot} \cong 2.6$ b for ^{9}Be and $E_n = 2.5$ MeV). All scattering and absorption processes are implicitly included in the MCNP5 simulation. Another consideration is the possibility of ‘cross-talk’ between the Zr and Be detectors, but this is found to be insignificant. Firstly, the ^{6}He β-particles are completely stopped by the Zr/Hf disc, and do not produce any pulses from the BGO detector. Secondly, Compton scattering of 2.319 and 2.189 MeV ^{90m}Zr γ-rays, will sometimes occur near the Be detector and an energetic electron can enter a proportional counter producing a cross-talk pulse from the Be detector. However, MCNP5 simulations show that <1% of ^{90m}Zr γ-rays will produce such cross-talk pulses. Moreover, the Be detector count is higher than the Zr detector count by a factor of ∼10 (see Fig. 5). Hence ^{90m}Zr γ-ray cross-talk pulses make a negligible contribution to the count from the Be detector.

3. Experimental procedure and methods

Measurements were made for deuterium gas pressures: 1.5 mbar and 2.0 to 10 mbar in steps of 1 mbar. Immediately following each shot, the pulses due to γ and β interactions in ZBP-0° and ZBP-90° were counted by the four scalers. On completion of the gate-interval, the values were recorded and the scalers reset, in readiness for the next shot. A natural γ-ray and cosmic-ray background rate is present for all four detectors. So periodically, background counts were collected over a 300 s interval. The four obtained background counts were scaled down (÷100) to match the 3.0 s gate-interval. Subtraction of background counts from the gross counts gave the four net count values, for each PF shot. Lastly, a correction factor was applied to include the expected number of counts occurring outside the 3.0 s gate-interval: the multiplying factor is $(\exp(-\lambda_j t_s)-\exp(-\lambda_j t_e))^{-1}$, where $t_s = 0.01$ s, $t_e = 3.01$ s and λ_j is 0.857 s^{-1} or 0.859 s^{-1}, for ^{90m}Zr or ^{6}He, respectively. (These are the same factors as contained in Eq. (5).) This correction increases net count values by about 9%. The corrected net count values are denoted C^0_{nZr}, C^0_{nBe}, C^{90}_{nZr}, C^{90}_{nBe}; their use in deriving neutron energies and anisotropies is discussed in Section 4.1.

After the first scan of deuterium gas pressures (10 values: 1.5–10 mbar), a second scan was performed for a subset pressures (5 values: 2–6 mbar) with a circular aluminium obstacle plate used to suppress the D$^+$ ion beam’s fusion contribution. This is similar to our previous NX2 beam-obstacle study [17] using a Pyrex obstacle. In the present work, an aluminium obstacle is used (instead of Pyrex) due to the more intense shock-wave resulting from the higher bank energy of the NX3 PF. The diameter of the aluminium plate is 21.0 cm, while the inner diameter of the NX3 vacuum chamber is 24.8 cm. The plate was positioned 6.0 cm from the anode tip — which approximates the centre

Table 1
Tabulation of number of shots fired over a range of deuterium gas pressure.

D_2 pressure (mbar)	w/o-OP		OP	
	# series	# shots	# series	# shots
1.5	2	32	–	–
2	2	32	2	32
3	3	48	3	48
4	4	64	3	48
5	2	28	2	32
6	2	32	3	48
7	4	56	–	–
8	2	32	–	–
9	2	32	–	–
10	2	32	–	–

of the plasma pinch. The anode tip to top plate distance is 27.6 cm. Hence, the presence of the obstacle reduces the D^+ beam path length to 22% of its obstacle-free value. Table 1 shows a summary of the data set. The abbreviations OP and w/o-OP indicate *with* and *without* obstacle plate, respectively. Typically, 16 shots were fired for each series with one gas-fill and one D_2 gas pressure. Shots were fired at about 1 min intervals. Excluding test shots, a total of 596 shots were performed for these experiments.

4. Results and discussion

4.1. Deriving experimental neutron energy and anisotropy values

For each PF shot, corrected net count values C^0_{nZr}, C^0_{nBe}, C^{90}_{nZr}, C^{90}_{nBe} are obtained as described in Section 3. Taken together with the MCNP5 simulation results, these experimental values enable the following experimental quantities to be calculated for each shot:

(I) effective neutron energy at $\theta = 0°$: denoted $E^{\mathrm{eff}}_{\mathrm{n}}(0°)$

(II) effective neutron energy at $\theta = 90°$: denoted $E^{\mathrm{eff}}_{\mathrm{n}}(90°)$

(III) neutron energy-anisotropy: $\Delta E_{\mathrm{n}} = E^{\mathrm{eff}}_{\mathrm{n}}(0°) - E^{\mathrm{eff}}_{\mathrm{n}}(90°)$

(IV & V) PSA neutron yields Y^0_{PSA} and Y^{90}_{PSA} (at 0° and 90°, respectively)

(VI) PSA neutron fluence-anisotropy $A_{\mathrm{nBe}} = Y^0_{\mathrm{PSA}}/Y^{90}_{\mathrm{PSA}}$.

The effective neutron energy $E^{\mathrm{eff}}_{\mathrm{n}}$ is obtained from interpolation from the curve in Fig. 5 (inputting either the $C^0_{\mathrm{Zr}}/C^0_{\mathrm{Be}}$ or $C^{90}_{\mathrm{Zr}}/C^{90}_{\mathrm{Be}}$ ratio). For neutron fluence anisotropy A_{nBe} we are using the same definition as [16], while for neutron energy anisotropy ΔE_{n} we are following [12].

To facilitate the discussion of neutron fluence-anisotropy for the w/o-OP versus OP cases, we will employ the simplifying assumption that all fusion neutrons originate at the centre of the plasma pinch (coincident with the centre of the anode tip): this is the Point Source Assumption (PSA). From the experimental results it will become clear that the PSA is wrong, and that neutrons are in fact emitted from an extended source. But the PSA serves a purpose in reaching that conclusion. PSA neutron yields are calculated from $Y^0_{\mathrm{PSA}} = C^0_{\mathrm{nBe}}/P_{\mathrm{Be}}\left(E^{\mathrm{eff}}_{\mathrm{n}}(0°)\right)$ and $Y^{90}_{\mathrm{PSA}} = C^{90}_{\mathrm{nBe}}/P_{\mathrm{Be}}\left(E^{\mathrm{eff}}_{\mathrm{n}}(90°)\right)$. For a typical shot: $Y^0_{\mathrm{PSA}} = C^0_{\mathrm{nBe}}/P_{\mathrm{Be}}(2.9\ \mathrm{MeV}) = C^0_{\mathrm{nBe}}/\left(5.493\times10^{-5}\right)$ and $Y^{90}_{\mathrm{PSA}} = \left(R_{\mathrm{Be90}}/R_{\mathrm{Be0}}\right)^2 C^{90}_{\mathrm{nBe}}/P_{\mathrm{Be}}(2.5\ \mathrm{MeV}) = 1.154C^{90}_{\mathrm{nBe}}/\left(4.427\times10^{-5}\right)$. Essentially Y^0_{PSA} and Y^{90}_{PSA} represent the neutron/sr at the ZBP detector, extended to 4π sr. Then the PSA neutron fluence-anisotropy is $A_{\mathrm{nBe}} = Y^0_{\mathrm{PSA}}/Y^{90}_{\mathrm{PSA}}$. For the typical shot above: $A_{\mathrm{nBe}} = 0.6983C^0_{\mathrm{nBe}}/C^{90}_{\mathrm{nBe}}$. Table 2 presents statistics for detector counts and derived quantities, for the 4.0 mbar D_2 gas pressure series (which has the largest number of performed shots = 64). It is seen that the Zr detector counts $C^{\theta}_{\mathrm{nZr}}$ are considerably lower than the corresponding Be detector counts $C^{\theta}_{\mathrm{nBe}}$. Hence, most of the statistical uncertainty in the $C^{\theta}_{\mathrm{Zr}}/C^{\theta}_{\mathrm{Be}}$ ratio (and consequently also in $E^{\mathrm{eff}}_{\mathrm{n}}$) arises from the $C^{\theta}_{\mathrm{nZr}}$ count value.

4.2. Shot-to-shot effective neutron energy and fluence anisotropy

Fig. 6 presents data for 28 consecutive PF shots at 5 mbar D_2 gas pressure, for the w/o-OP case only. Fig. 6(a) shows plots of C^{90}_{nBe} count and neutron fluence-anisotropy A_{nBe} for each shot. Error bars represent the standard error from the statistics of counting, but not systematic uncertainties, which are discussed in Section 2.3. The experimental A_{nBe} values are above those expected $(1.0 < A < 1.5)$ from DD fusion occurring in a small beam–target source (i.e. the plasma pinch column) [26–28]. Also, there are large shot to shot variations in measured A_{nBe} values, ranging from ∼2.6 to ∼4.6. We consider that the large A_{nBe} value are due to the forward-cone of beam–target fusion, which shifts the neutron emission centre-of-gravity (n-CG) towards the ZBP-0° detector. To be more precise, the fast deuteron beam population $f\left(E_d,\theta_d\right)$ is an energy and angular distribution that is assumed to be azimuthally symmetric. The large variations in A_{nBe} reflect variations in the relative contributions of forward-cone versus plasma pinch zone fusion (i.e. the 'strength' of the D^+ ion beam varies from shot to shot). The lowest A_{nBe} values will occur for shots where the DD fusion is mostly confined to the pinch zone. C^{90}_{nBe} is much less influenced by the n-CG shift than C^0_{nBe}. Therefore, C^{90}_{nBe} is used in Fig. 6(a) as a quantity that is first-order proportional to the neutron yield for the PF shot. Comparison of the two data in Fig. 6(a), does indicate a slight positive correlation between A_{nBe} and neutron yield per shot. In addition, the A_{nBe} plot in Fig. 6(a) can be compared with the A_{nBe} plots shown in Fig. 4 of Ref. [17] (for the NX2 PF): in which the 'without Pyrex' A_{nBe} plot corresponds to the w/o-OP case of the present work. The generally lower values of A_{nBe} reported in [17] are most likely due to the smaller anode-tip to top plate distance of 11.8 cm for the NX2 PF, versus 27.6 cm for the NX3 PF. Also in [17], plots in Fig. 4 show wide shot-to-shot A_{nBe} fluctuations (similar to the present data), and mean values $\langle A_{\mathrm{nBe}}\rangle$ decreasing as the pinch to obstacle distance decreases.

For the same series of 28 shots, Fig. 6(b) shows the effective neutron energies $E^{\mathrm{eff}}_{\mathrm{n}}$ at 0° and 90°. Per shot $E^{\mathrm{eff}}_{\mathrm{n}}(90°)$ values are mostly in the range 2.45 to 2.55 MeV, with mean of $\left\langle E^{\mathrm{eff}}_{\mathrm{n}}(90°)\right\rangle = 2.50$ MeV. The corresponding 0° values $E^{\mathrm{eff}}_{\mathrm{n}}(0°)$ are mostly above 2.8 MeV, with mean of $\left\langle E^{\mathrm{eff}}_{\mathrm{n}}(0°)\right\rangle = 2.81$ MeV. As would be expected for beam–target fusion $E^{\mathrm{eff}}_{\mathrm{n}}(0°) > E^{\mathrm{eff}}_{\mathrm{n}}(90°)$ for every shot. The average neutron energy anisotropy value for the series is $\left\langle \Delta E^{\mathrm{eff}}_{\mathrm{n}}\right\rangle = \left\langle E^{\mathrm{eff}}_{\mathrm{n}}(0°) - E^{\mathrm{eff}}_{\mathrm{n}}(90°)\right\rangle = 0.31$ MeV. (Whereas thermonuclear fusion would be characterized by: $E^{\mathrm{eff}}_{\mathrm{n}}(0°) = E^{\mathrm{eff}}_{\mathrm{n}}(90°) = 2.45$ MeV, and $\Delta E^{\mathrm{eff}}_{\mathrm{n}} = 0$ MeV). Hence, Fig. 6 provides clear evidence that beam–target fusion is dominant, and it occurs throughout an extended forward cone where the cold deuterium gas is bombarded by energetic D^+ ions emitted from the pinch. In terms of DD reaction kinematics, 2.81 MeV neutrons at 0° are produced by a narrow pencil beam $\theta \approx 0°$ of D^+ ions with $E_d = 82$ keV reacting with stationary deuterons (from Eq. (1)); a broader conical D^+ beam clearly requires $E_d > 82$ keV. Fig. 6(b) shows that there is significant variation in neutron energy from shot to shot, and the variance is larger for $E^{\mathrm{eff}}_{\mathrm{n}}(0°)$ than it is for $E^{\mathrm{eff}}_{\mathrm{n}}(90°)$. Later, Fig. 7(b), shows that similar energy anisotropy values $\Delta E^{\mathrm{eff}}_{\mathrm{n}} \sim 0.3$ MeV are observed at other D_2 gas pressures, for both the OP and w/o-OP cases.

4.3. Average neutron yield vs. D_2 gas pressure

From the data for all 596 shots, values of mean neutron yield $Y^{\theta}_{\mathrm{PSA}}$, mean neutron energy $E^{\mathrm{eff}}_{\mathrm{n}}(\theta)$ and mean neutron fluence-anisotropy A_{nBe}, are calculated using all the data at each D_2 gas pressure (grouped by $\theta = 0°\&90°$, and w/o-OP and OP cases). Fig. 7(a–c) presents plots of these mean values $\left\langle Y^{\theta}_{\mathrm{PSA}}\right\rangle$, $\left\langle E^{\mathrm{eff}}_{\mathrm{n}}(\theta)\right\rangle$ and $\langle A_{\mathrm{nBe}}\rangle$ as a function of D_2 gas pressure.

Fig. 7(a) shows plots of $\left\langle Y^0_{\mathrm{PSA}}\right\rangle$ and $\left\langle Y^{90}_{\mathrm{PSA}}\right\rangle$ for the w/o-OP and OP cases. For w/o-OP: $\langle Y_{\mathrm{PSA}}\rangle$ forms a broad peak with its maximum around 5 mbar. Whereas, for OP: $\langle Y_{\mathrm{PSA}}\rangle$ is considerably lower and increases gradually with D_2 gas pressure up to 6 mbar. It is evident from Fig. 7(a) that the 0° orientation $\langle Y_{\mathrm{PSA}}\rangle$ OP values are much lower than the

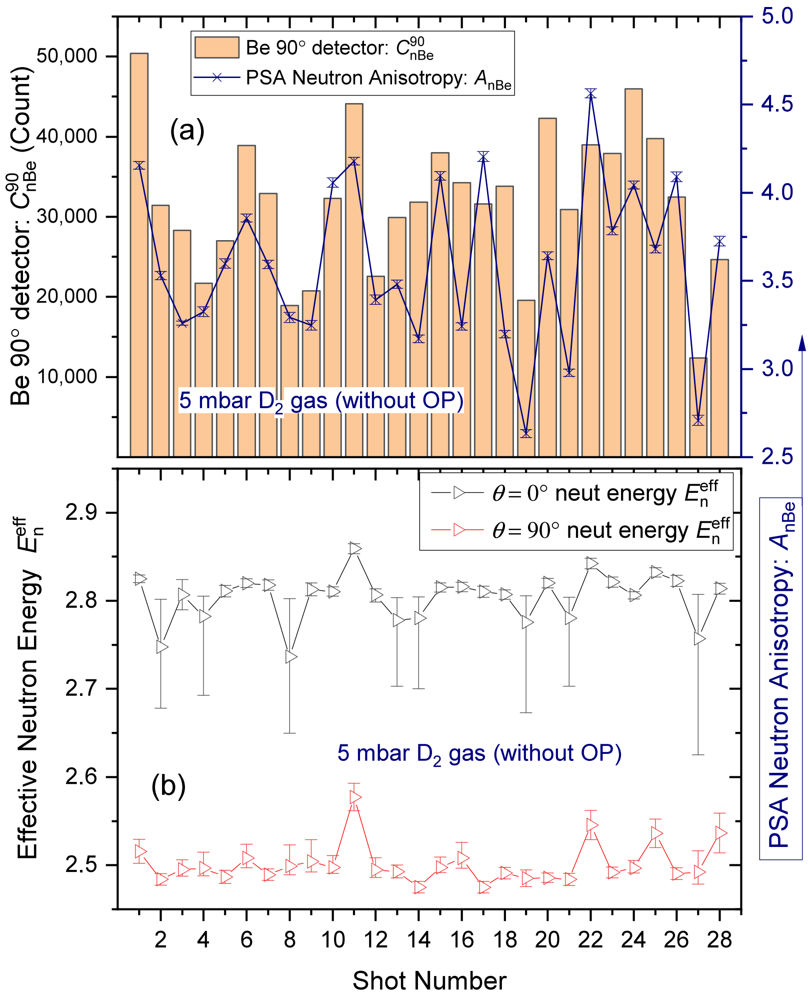

Fig. 6. Data for 2 series of (12 + 16) shots at 5 mbar D_2 gas pressure without the aluminium plate obstacle (w/o-OP). Error bars represent the standard error from the statistics of counting. (a) Shot-to-shot Be-90° count (bars; left axis) and neutron fluence anisotropy (line; right axis). (b) Shot-to-shot neutron energy at $\theta = 0°$ (black) and 90° detectors (red).

corresponding $\langle Y_{PSA} \rangle$ w/o-OP values; i.e. introduction of the obstacle reduces the number of neutrons observed at 0° by a large factor. There is also a reduction in $\langle Y_{PSA} \rangle$ at 90° for OP vs. w/o-OP, although the reduction factor is significantly smaller. These observations are entirely consistent with the idea that a forwardly-directed beam of energetic D^+ ions emitted from the pinch produces fusion throughout a conical volume (apex at plasma pinch, base at chamber top plate). In effect, the obstacle plate cuts a frustum from the fusion-cone, and the centre-of-gravity of this frustum is much closer to the 0° detector than it is to the 90° detector. For the OP case, the cone height is reduced to 22% of its w/o-OP value.

Table 2 summarizes all the 4.0 mbar D_2 gas pressure data, showing average corrected net count for Zr and Be detectors, grouped by 0° or 90° orientation, and w/o-OP or OP cases, and other derived quantities. These data have comparable statistics to those of Fig. 6, which presents plots for all 5 mbar data. Comparing the w/o-OP and OP cases at 0° in Table 2, the Zr and Be counts are reduced by factors of 5.1 and 5.9 when the beam-obstacle is inserted. The corresponding reduction factors at 90°, are considerably lower, at 2.2 and 2.7. Data at other D_2 gas pressures show similar trends. Also $\left\langle Y^0_{PSA} \right\rangle$ is reduced by a significantly larger factor than $\left\langle Y^{90}_{PSA} \right\rangle$ when the obstacle plate is inserted. These data, along with those at other pressures, are consistent with the interpretation that the obstacle plate cuts a frustum from the cone of fusion produced by forwardly-directed energetic D^+ ions; and this frustum represents a substantial fraction of the (w/o-OP) neutron yield per shot (in PSA terms: it contributes ~75% of the w/o-OP yield).

4.4. Average neutron fluence anisotropy vs. D_2 gas pressure

For an accelerator-like DD neutron source (i.e. narrow mono-directional mono-energetic ($= E_d$) D^+ ion beam incident on a thin deuterated target) the pure fluence-anisotropy value $A_{n0,90}$ is a function of E_d only. A_{n0} is calculable from the $^2H(d,n)^3He$ differential cross-section $(d\sigma/d\Omega)_{DD}\,(\theta_n)$ [27] and CM-to-Lab frame transformation [5]. Example values are: $A_{n0,90} = 1.77$ and 2.52, for $E_d = 30$ and 100 keV, respectively. For the plasma focus situation, the fluence-anisotropy is modified considerably by the angular θ_d and energy E_d distributions of the beam deuterons. A modified anisotropy A_n can be calculated for a given $f\left(E_d, \theta_d\right)$ [6,14], and typically $A_n < 1.5$. But further modification of fluence anisotropy results from the spatial extent of the fusion source and detector distance from the extended source [28]. For thermalized (moderated) neutron detectors the neutron 'room return' is also significant [29]. For the NX3 PF experimental situation, it is necessary to position the ZBP detectors close to the pinch (i.e. immediately outside the vacuum chamber) to obtain sufficiently high Zr detector counts (C^0_{nZr} and C^{90}_{nZr}), that $E^{eff}_n(0°)$ and $E^{eff}_n(90°)$ values of good accuracy are obtained for individual PF shots. The measured A_{nBe} values are therefore necessarily 'near field' values. Consequently, the measured A_{nBe} values provide little information on E_d, but they do give valuable information on the axial extent of the neutron source.

Fig. 7(c) shows a plot of mean neutron fluence-anisotropy $\langle A_{nBe} \rangle$ versus D_2 gas pressure, for the two OP cases. For the unobstructed beam (w/o-OP) case, the $\langle A_{nBe} \rangle$ values are in the range 1.5 to 4.0, which lie generally above the pure $A_n < 1.5$ expected for accelerator-like DD

Table 2
For all 4.0 mbar D_2 data: tabulation of values for ZBP-0° and ZBP-90° activation detectors, plus the two obstacle-plate cases (w/o-OP and OP). Data for each case, is: number of shots fired, mean corrected detector counts $\langle C^{\theta}_{\mathrm{nZr}} \rangle$ and $\langle C^{\theta}_{\mathrm{nBe}} \rangle$, and other derived quantities.

4 mbar D_2 gas data (all shots)	w/o-OP 0°	w/o-OP 90°	OP 0°	OP 90°
# shots	64 shots		48 shots	
$\langle C^{\theta}_{\mathrm{nZr}} \rangle$	2,634	300.3	518.1	134.4
$\langle C^{\theta}_{\mathrm{nBe}} \rangle$	75,404	17,090	12,870	6,312
$\langle Y^{\theta}_{\mathrm{PSA}} \rangle$	1.37×10^9	0.449×10^9	2.34×10^8	1.57×10^8
$\max(Y^{\theta}_{\mathrm{PSA}})$	4.05×10^9	1.18×10^9	9.26×10^8	5.31×10^8
$\langle E^{\mathrm{eff}}_{\mathrm{n}} \rangle(\theta)$	2.82 MeV	2.49 MeV	2.83 MeV	2.55 MeV
$\langle \Delta E^{\mathrm{eff}}_{\mathrm{n}} \rangle$	0.33 MeV		0.28 MeV	
$\langle A_{\mathrm{nBe}} \rangle$	3.06		1.49	

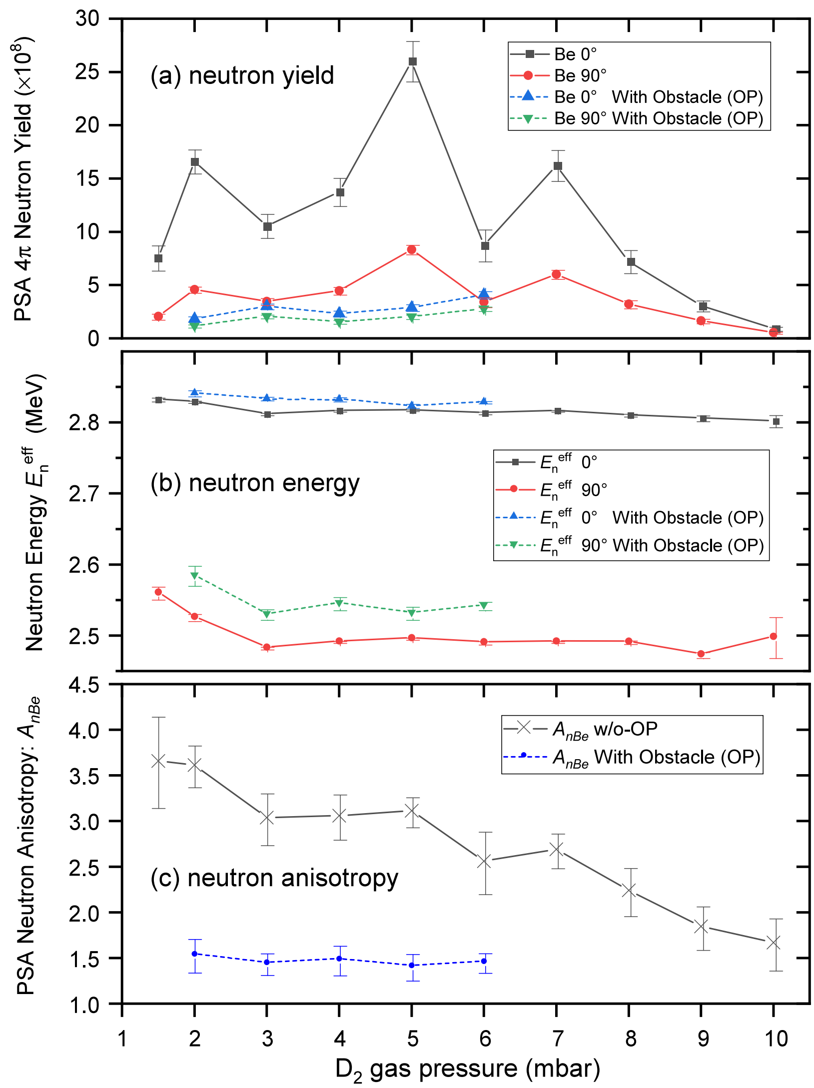

Fig. 7. Plotted against D_2 gas pressure are: (a) PSA neutron yield (0° & 90°), (b) Average effective neutron energies (0° & 90°), and (c) Average neutron anisotropy. Data for w/o-OP or OP cases are displayed with a solid or dashed line, respectively. Error bars represent the standard error from the statistics of counting.

reaction conditions. These $\langle A_{\mathrm{nBe}} \rangle$ anisotropy values highlight that the 'centre of gravity' of neutron emission is located at a forward position, between the pinch and top plate. By contrast, for the obstructed beam (OP) case, the observed $\langle A_{\mathrm{nBe}} \rangle$ anisotropy values are all close to 1.5 which is more consistent with (far field) anisotropy values expected purely from DD beam–target kinematics [26–28].

Another aspect of the data displayed in Fig. 7(c) is the trend of the average neutron fluence-anisotropy $\langle A_{\mathrm{nBe}} \rangle$ with D_2 gas pressure. For the w/o-OP case, $\langle A_{\mathrm{nBe}} \rangle$ decreases steadily with increasing pressure. This $\langle A_{\mathrm{nBe}} \rangle$ trend is similar to that observed in Ref. [17] Fig. 3 (top left pane) for the NX2 PF. In contrast to the w/o-OP data, for the OP case $\langle A_{\mathrm{nBe}} \rangle$ remains almost constant with gas pressure — at least up to 6 mbar, the highest pressure for which the OP case was studied. A simplistic interpretation would be that the D^+ ion beam is 'stronger' at lower gas pressures. However, in contrast to the $\langle A_{nBe} \rangle$ trend, Fig. 7(b) shows that the average neutron energy $\langle E_n^{eff}(0°) \rangle$ is only slightly affected by gas pressure, for both the OP and w/o-OP cases. These observations taken together suggest that the fast-deuteron energy spectrum is relatively unaffected by gas pressure. Many PF ion beam studies have found beam deuteron energy spectra of the form $dN/dE \cong AE_d^{-n}$. The present results suggest that with increasing gas pressure, the factor A decreases, but n is relatively unaffected.

4.5. Average neutron energy vs. D_2 gas pressure

Fig. 7(b) shows plots of $\langle E_n^{eff}(0°) \rangle$ and $\langle E_n^{eff}(90°) \rangle$ for the w/o-OP and OP cases. It is seen that the average neutron energy $\langle E_n^{eff} \rangle$ depends only weakly on D_2 gas pressure. Comparing the w/o-OP and OP plots shows that insertion of the obstacle plate gives a small, but persistent, increase in measured $\langle E_n^{eff} \rangle$ values over all pressures. The OP-related increase in $\langle E_n^{eff} \rangle$ is larger for the 90° direction. Our interpretation is that $E_n^{eff}(90°)$ increases with OP insertion because the OP cuts the downstream frustum of beam–target fusion (due most likely to 30 keV < E_d < 100 keV deuterons): consequently, many of the larger θ angle ($90° < \theta \leq 180°$) $^2H(d,n)^3He$ reactions are eliminated from the ZBP-90° viewpoint. Eq. (1) (and Fig. 1) dictates that neutron energy E_n decreases as θ increases. Hence, for the OP case, the ZBP-90° detector will view generally $\theta \approx 90°$ fusion, and $\langle E_n^{eff} \rangle$ is therefore higher than it is for the w/o-OP case, where ZBP-90° views the whole fusion cone, for which $90° < \theta \leq 180°$.

A similar interpretation applies to the small increases in $E_n^{eff}(0°)$ with obstacle insertion. Now, for the OP case, the ZBP-0° detector will view mainly small angle ($\theta \approx 0°$) $^2H(d,n)^3He$ reactions occurring in the pinch zone. By contrast, in the w/o-OP case, the ZBP-0° detector will be viewing fusion from the whole cone, and the forward frustum will contain many larger θ angle fusion reactions: where neutrons emerge with $0° \leq \theta < 90°$. Therefore, again $\langle E_n^{eff} \rangle$ should be higher for the OP case than it is for the w/o-OP case, as is observed experimentally. In conclusion, Fig. 7(b) plots of $\langle E_n^{eff}(0°) \rangle$ and $\langle E_n^{eff}(90°) \rangle$ for w/o-OP and OP cases, are wholly explicable in terms of beam–target fusion occurring throughout an extended cone around the forward PF axis, due to an intense beam of energetic D^+ ions emitted from the plasma pinch.

The small change in $\langle E_n^{eff}(0°) \rangle$ energy values for the OP vs. w/o-OP cases has another implication. The fusion originating within the pinch and in the forward cone both have $\Delta E_n \approx 0.3$ MeV. This result is inconsistent with the Gyrating Particle Model (GPM) [18], in which fusion is produced by D^+ ions trapped in gyrating-particle orbits by the strong azimuthal magnetic field B_ϕ within, and around, the plasma pinch. The widely distributed velocity-directions of these trapped D^+ ions would lead to a nearly isotropic angular distribution of $^2H(d,n)^3He$ neutrons. Moreover, the energy distribution of these GPM neutrons would be centred around 2.5 MeV. Hence the GPM leads to the expectation that $\langle E_n^{eff}(0°) \rangle$ should fall for the OP case because fusion originating within the plasma pinch represents a much larger fraction of the neutron yield than it does for the w/o-OP case. The present experimental results are, however, clearly contrary to the above expectation: $\langle E_n^{eff}(0°) \rangle$ increases slightly with OP insertion. This happens despite the large drop in $\langle Y_{PSA}^0 \rangle$ which shows that a large beam–target fusion contribution (i.e. from the frustum) has been eliminated by OP insertion. We therefore conclude that the GPM makes no significant contribution to fusion in the NX3 PF.

5. Conclusions

The dissimilar energy dependence of the fast neutron activation cross-sections for Zr and Be, $\sigma_{Zr}^a(E_n)$ and $\sigma_{Be}^a(E_n)$, enables a (quasi-average) effective neutron energy E_n^{eff} to be measured using a Zr–Be detector Pair (ZBP). The method is applicable to pulsed DD fusion devices, and the range of neutron energies relevant to thermonuclear and beam–target fusion. We have demonstrated that for a deuterium plasma focus with a yield of ~10^9 neutron/shot, that Zr–Be detectors can be used to measure shot-to-shot neutron energies (E_n^{eff}) at different θ angles, simultaneously. Two ZBP detectors have been constructed, and the necessary MCNP5 Monte Carlo simulations have been carried out, so that neutron energy, yield and anisotropy measurements can be made for many individual shots of the NX3 PF operated in D_2 gas. The ZBP detectors were placed at $\theta = 0°$ and 90° wrt the PF axis, so that neutron energies, $E_n^{eff}(0°)$ and $E_n^{eff}(90°)$, were obtained for each shot. Plots of $E_n^{eff}(0°)$ and $E_n^{eff}(90°)$ vs. shot number in Fig. 6(b), indicate that neutron energy varies significantly from shot-to-shot at 0° orientation, but much less so at 90°. Fig. 7(b) shows that varying D_2 gas pressures has little effect on the energies of emitted neutrons.

Moreover, 'near field' neutron yield and fluence-anisotropy values (Y_{PSA}^0, Y_{PSA}^{90}, and $A_{nBe} = Y_{PSA}^0/Y_{PSA}^{90}$) are also obtained for each shot, and these yield information on the axial extent of neutron emission within the NX3 PF device. Measurements were made for a series of 388 shots over the range of D_2 gas pressures 1.5 to 10 mbar (w/o-OP). Then, in order to investigate the contribution to fusion yield, and neutron energy, of the conical D^+ ion beam versus pinch zone, an aluminium obstacle plate (OP) was positioned 6.0 cm from the anode tip, and a second series of 208 shots were performed over a smaller range of D_2gas pressures, 2 to 6 mbar.

If thermonuclear fusion localized in the hot plasma pinch were responsible for the bulk of neutron emission in the deuterium PF, then it would be characterized by: $E_n^{eff}(0°) = E_n^{eff}(90°) = 2.45$ MeV, $\Delta E_n^{eff} = 0.0$ MeV, and $A_{nBe} = 1.0$. Importantly, these values would be unaffected by insertion of the beam obstacle. Also the PSA neutron yields Y_{PSA}^0 and Y_{PSA}^{90} would be the same for the w/o-OP versus OP cases. The collected experimental data conspicuously contradict the expectations of the thermonuclear fusion model. We therefore conclude that any thermonuclear contribution to the neutron yield is negligible.

PSA neutron yields Y_{PSA}^0 and Y_{PSA}^{90} decrease markedly for OP insertion, demonstrating that the forward (6 cm < z < z_{tp}) conical D^+ beam contributes a large fraction of the yield. Insertion of OP cuts a frustum from the forward fusion-cone (6 cm < z < z_{tp}), and the centre-of-gravity of this frustum is much closer to the 0° detector than it is to the 90° detector. The z < 6 cm region, which includes the pinch column, contributes the remainder of the yield. Due to the fusion centre-of-gravity shift between the w/o-OP and OP cases, Y_{PSA}^{90} is a better first-order approximation to total neutron yield than Y_{PSA}^0. For the peak neutron yield pressure of 5 mbar, $\langle Y_{PSA}^{90} \rangle$(OP) is ~25% of $\langle Y_{PSA}^{90} \rangle$(w/o-OP) indicating that the (6 cm < z < z_{tp}) conical D^+ beam contributes an apparent (in PSA terms) ~75% of the fusion yield. Hence, the PF is a spatially extended (conical) source of neutrons, due to a fast deuteron beam population $f(E_d, \theta_d)$ emitted from the pinch.

Mean neutron fluence-anisotropy $\langle A_{nBe} = Y_{PSA}^0/Y_{PSA}^{90} \rangle$ also decreases markedly for OP insertion. For w/o-OP case, neutron fluence-anisotropy $\langle A_{nBe} \rangle$ is larger than pure A_n values, because the neutron emission centre-of-gravity (n-CG) is located on the forward PF axis. With OP insertion, $\langle A_{nBe} \rangle$ values reduce from ~3.5 to ~1.5. The reductions in Y_{PSA}^0 and A_{nBe} for OP relative to w/o-OP are strong evidence of the large fusion contribution from 'cone zone' fusion. The trend of decreasing $\langle A_{nBe} \rangle$ with D_2 gas pressure (Fig. 7(c), w/o-OP) accords with trend observed in [17]. Also, the $\langle A_{nBe} \rangle$ reduction with OP insertion resembles that observed in Ref. [17], Fig. 3. These $\langle A_{nBe} \rangle$ trends indicate consistent behaviour between the NX2 and NX3 PF devices with capacitor bank energies (E_{CB}) of 1.6 and 7.2 kJ, respectively.

The Gyrating Particle Model (GPM) is inconsistent with the observed neutron energy values $E_n^{eff}(0°)$ and $E_n^{eff}(90°)$ for essentially all shots, over the range of gas pressures studied. Across the dataset, $E_n^{eff}(0°)$ and $E_n^{eff}(90°)$ remain consistent with a straightforward beam–target model of PF fusion. The GPM may become significant for larger PF devices, as both plasma pinch diameter and pinch current (and therefore azimuthal magnetic strength B_ϕ) increase with PF capacitor bank energy $\left(E_{\mathrm{CB}}\right)$. But the GPM is inapplicable for the NX3 PF, and probably for other PF devices with $E_{\mathrm{CB}} < 10\mathrm{kJ}$.

Deuterium gas pressure has only a minor effect on the energy of the emitted DD neutrons. $\left\langle E_{\mathrm{n}}^{\mathrm{eff}}(0°)\right\rangle$ values are close to 2.82 MeV (`w/o-OP`), but display a very gradual decrease with increasing gas pressure. $\left\langle E_{\mathrm{n}}^{\mathrm{eff}}(90°)\right\rangle$ values are close to 2.50 MeV (`w/o-OP`), 2.55 MeV (`OP`) with some increase at the lowest pressures. Interestingly, $\left\langle E_{\mathrm{n}}^{\mathrm{eff}}(0°)\right\rangle$ and $\left\langle E_{\mathrm{n}}^{\mathrm{eff}}(90°)\right\rangle$ values are higher for the `OP` case than for the `w/o-OP` case. As discussed in Section 4.5, this observation is also entirely consistent with beam–target fusion being dominant — in both the pinch and the forward cone.

If detailed theoretical models of the energy spectrum and directional distribution of fast-deuterons, $f\left(E_d, \theta_d\right)$, emitted from the plasma pinch become available, then the expected values of $E_{\mathrm{n}}^{\mathrm{eff}}(\theta)$, $Y_{\mathrm{PSA}}^{\theta}$ and A_{nBe} can be calculated and compared with these experimental measurements.

Finally, Zr–Be FNA detectors could be employed for experiments on other pulsed non-thermal fusion devices such as Z-pinches and femto/pico-second laser deuterated-target experiments. The short ∼0.8 s half-lives of the two activated products ($^{90\mathrm{m}}$Zr and ^{6}He) make Zr–Be FNA detectors suitable for experiments with relatively high repetition rates up to ∼0.2 Hz. The most direct way of overcoming the systematic uncertainties in $E_{\mathrm{n}}^{\mathrm{eff}}$ which arise due to the uncertainties in the activation cross-sections $\sigma_{\mathrm{Zr}}^{\mathrm{a}}(E_{\mathrm{n}})$ and $\sigma_{\mathrm{Be}}^{\mathrm{a}}(E_{\mathrm{n}})$, would be to energy calibrate a specific design of Zr/Be FNA detector-pair using an accelerator-based neutron source, with a similar experimental arrangement as in Ref. [22].

CRediT authorship contribution statement

S.V. Springham: Conceptualization, Methodology, Formal analysis, Investigation, Resources, Writing - review & editing, Project administration, Supervision. **R. Verma:** Resources, Conceptualization, Formal analysis, Investigation. **M.S.N. Zaw:** Investigation, Data curation. **R.S. Rawat:** Funding acquisition, Resources. **P. Lee:** Resources, Editing. **A. Talebitaher:** Investigation, Supervision. **J.H. Ang:** Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The authors are grateful to the National Institute of Education, Nanyang Technological University, Singapore, for AcRF grant number RI 7/16 SSV.

References

[1] A. Bernard, et al., Phys. Fluids 18 (1975) 180–194.
[2] A. Bernard, et al., Nucl. Instrum. Methods 145 (1977) 191–218.
[3] S.V. Springham, S. Lee, et al., Plasma Phys. Control. Fusion 42 (2000) 1023–1032.
[4] J.H. Lee, et al., Phys. Fluids 14 (1971) 2217–2223.
[5] K.S. Krane, Introductory Nuclear Physics, second ed., John Wiley & Sons, Inc., New York, USA, 1988, p. 382.
[6] I. Tiseanu, N. Mandache, V. Zambreanu, Plasma Phys. Control. Fusion 36 (1994) 417–432.
[7] IAEA Nuclear Data Services (http://www-nds.iaea.org/exfor/).
[8] M.M. Milanese, J.O. Pouzo, Nucl. Fusion 18 (1978) 533–536.
[9] H. Conrads, et al., Phys. Fluids 15 (1972) 209–211.
[10] H. Rapp, Phys. Lett. 57A (1976) 46–48.
[11] C. Patou, A. Simonnet, J.P. Watteau, Phys. Lett. A 29 (1969) 1–2.
[12] M.J. Bernstein, D.A. Meskan, H.L.L. van Paassen, Phys. Fluids 12 (1969) 2193–2202.
[13] M.J. Bernstein, F. Hai, Phys. Lett. 31A (1970) 317–318.
[14] M.J. Bernstein, G.G. Comisar, Phys. Fluids 15 (1972) 700–707.
[15] G.F. Knoll, Radiation Detection and Measurement, third ed., John Wiley & Sons, Inc., New York, USA, 2000, p. 705 & 749.
[16] A. Talebitaher, et al., Nucl. Instrum. Methods Phys. Res. A 659 (2011) 361–367.
[17] A. Talebitaher, S.V. Springham, R.S. Rawat, P. Lee, Nucl. Instrum. Methods Phys. Res. A 848 (2017) 60–65.
[18] U. Jäger, H. Herold, Nucl. Fusion 27 (1987) 407–423.
[19] C.L. Ruiz, et al., Phys. Rev. Accel. Beams 22 (2019) 042901.
[20] P.H. Stelson, E.C. Campbell, Phys. Rev. 106 (1957) 1252–1255.
[21] National Nuclear Data Center (NNDC), (http://www.nndc.bnl.gov).
[22] T. Shimizu, et al., Ann. Nucl. Energy 32 (2005) 949–963.
[23] R. Verma, et al., IEEE Trans. Plasma Sci. 40 (2012) 3280–3289.
[24] T. Shimizu, et al., Ann. Nucl. Energy 31 (2004) 1883–1900.
[25] M.J. Berger, et al., NIST XCOM: Photon cross section database (vers. 1.5), 2010, (http://physics.nist.gov/xcom), NIST, USA.
[26] J.H. Lee, L.P. Shomo, K.H. Kim, Phys. Fluids 15 (1972) 2433–2438.
[27] R.E. Brown, N. Jarmie, Phys. Rev. C 41 (1990) 1391–1399.
[28] I. Tiseanu, N. Mandache, AIP Conf. Proc. 299 (1994) 356–362, http://dx.doi.org/10.1063/1.2949177.
[29] M. Frignani, et al., Radn. Prot. Dosim. 115 (2005) 380–385.